Yiming Meng

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Education

Northeastern University, China Sep 2022 - Jul 2026

B.Eng. in Robotics Engineering, School of Robotics Science and Engineering

- GPA: 4.17/5.00; **major rank: 1/72.**
- **National Scholarship (twice);** Outstanding Graduate of Liaoning Province.

Research Experience

Coordi-π (First Author) Apr 2026 - Aug 2026

Shanghai Jiao Tong University, ScaleLab | Research Assistant | Mentor: Yao Mu

Coordi-π: Coordination-Aware Dual-hand Action Representation for Bimanual Dexterous Manipulation | project website

- Coordi-π models hand identity and temporal coordination in a bimanual VLA for dexterous manipulation.
- Designed interleaved arm-hand tokens, hand-identity embeddings, and factorized rotary positional encoding. Proposed a bimanual coordination metric for role allocation and temporal dependencies, and evaluated the model through ablations.
- Highest mean success among evaluated DexJoCo methods; improved over π0.5 by **19.4 percentage points** on five simulation tasks and **13.7** on four real-world tasks. **First author; submitted to ICRA 2027, under review.**

PhysicalRSI (Co-contributor) Aug 2026 - Present

KAI & HKU MMLab | Research Intern | Mentor: Ping Luo

PhysicalRSI 1.0: Recursive Self-Harness for Scaling Embodied Skills | project website

- PhysicalRSI uses multimodal reasoning to plan robot tasks and executes them through code policies or VLA skills. Execution feedback and sparse memory guide revisions to planning and control routines across trials.
- Contributed to skill selection by matching instructions and visual observations to skill capabilities and constraints; helped design code policies for object localization, motion planning, and gripper control, and led the real-robot experiments.
- **PhysicalRSI ranked No. 1 on RoboDojo-Sim** (Sep 28, 2026).

X-VLA (Contributor) Dec 2024 - Mar 2025

Tsinghua University, AIR | Research Assistant | Mentor: Xianyuan Zhan

X-VLA: Soft-Prompted Transformer as Scalable Cross-Embodiment Vision-Language-Action Model | project website

- X-VLA combines embodiment-specific soft prompts with flow matching for cross-embodiment manipulation.
- Contributed to X-VLA's simulation evaluation: surveyed coverage and protocols for benchmark selection; deployed LIBERO and SimplerEnv, validated policy inference and measured task success rates.
- **X-VLA won the AgiBot World Challenge at IROS 2025.**

Competitions

National University Student Intelligent Car Competition (Outdoor)	**National Second Prize**
RoboCom Robot Developer Competition, Virtual Simulation Division	**National Second Prize**
China Robot Competition and RoboCup China, FIRA Division	**National Second Prize**
National English Competition for College Students	National Third Prize
Mathematical Contest in Modeling (MCM)	International Second Prize
CUMCM (Mathematical Modeling)	Provincial First Prize
National College Mathematics Competition	Provincial First Prize

Honors and Scholarships

- **National Scholarship (twice)**
- University scholarships (twice each): First-Class Outstanding Student, Excellence, and Future Technology.
- University honors: Outstanding Student Role Model (twice); Outstanding Student Leader.

Skills and Languages

- Programming: Python, C++, MATLAB | English: CET-6 596.